Cognition as a positional good

The planning horizon produces nothing — and discounting is what keeps the race from turning into hallucination

(preprint — English translation of `papers/paper-2-bem-posicional.md`; source of record remains the Portuguese original, including all inline addenda)

Abstract

In a minimal, deterministic simulated world, agent-blocks plan `h` ticks ahead, discounting the future by `δ`, and both parameters are heritable traits with mutation. We ask when it pays to think. The answer has three layers, and each one overturned the previous one.

First: planning deep is **individually advantageous and collectively costly**. Equilibrium population is maximal at `h = 1` and falls monotonically with depth; and in an invasion trial the deep planner **displaces** the shallow one. Cognitive horizon is a **positional good** (Hirsch, 1976) — a Red Queen arms race (Van Valen, 1973).

Second: we measured *how much* of a positional good it is. It is **pure**. In a single-type population, with no rival of another type, every planning step past the second makes the block **harvest worse**: with `h = 12`, +20 units of standing food are left over versus `h = 1` (`t = +12.5`). The horizon **produces nothing**. Everything it buys is taken from a neighbor.

Third, and this is the one that surprised us: the race **has an endogenous brake**, and the brake is not the price of cognition — it is the **imprecision of the prediction itself**. Discounting is not impatience: it is a **regularizer**. It decides how much weight the plan's hallucinated tail receives ($δ^k$). At `δ = 0.80` the tail is discounted into irrelevance and the extra depth is harmless; at `δ = 0.95` it enters with half its weight, and there planning deep is *absolutely* bad — a single-type population with `h = 12` sustains 5.5 fewer blocks than the same population with `h = 8` (`t = −4.3`), with no one there to take its food. The damage shows a dose-response in `δ` and **more than triples** from 0.80 to 0.95.

From this follows a counter-intuitive result we report as the strongest of the paper: **more patience ⇒ a shallower optimal horizon.** The more weight is placed on the distant future, the less far one can look without being poisoned by one's own prediction.

And from this follows the boundary of the thesis: the horizon is a positional good **where discounting protects it**. Above that, it is not an expensive good — it is a **mistake**, and selection corrects it.

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1. The question

"Evolution learns to plan farther ahead" is a common narrative, and in this world it is ill-formed in two senses.

The first is metrological: `horizon` **is not identifiable on its own**. A block with horizon 11 and discount 0.66 weighs the 3rd step at `0.66³ ≈ 0.29` and the 11th at `0.01`: it *declares* a deep horizon and **thinks shallow**. The two traits are compensatory — `corr(horizon_mean, discount_mean)` = −0.93 / −0.46 / −0.72 in the late regime — and any claim about planning depth needs the **pair** `(h, δ)`. Much of the between-seed divergence that fooled us through three hypotheses was an artifact of looking at a non-identifiable trait in isolation.

The second is substantive, and it is this paper's subject: in the part where the narrative is true, **it is not the good news it looks like.**

2. The apparatus, and why a toy

A single ~56 KB C file, libc only. Blocks occupy a grid, eat procedural food patches, spend energy to exist, and reproduce. Each block decides by looking **only at its 3×3 neighborhood**; none reads global state. Four traits govern the decision (`urgency`, `space_weight`, `discount`, `horizon`), inherited with mutation.

The property that makes the apparatus useful is **total determinism**: the entire universe is `f(seed)`. This buys what a real system does not give — **pinning a trait** across the whole population and re-running the *same* world; running **invasion trials** with exact inheritance, where the trait mean in the log *is* the frequency; and comparing conditions without the world shifting underneath.

3. The positional good (the first layer)

Group landscape. Fixing `horizon = h` across the whole population, equilibrium population falls **monotonically**:

h	1	2	3	4	6	8	10	12
pop	**295.9**	295.0	293.6	292.7	290.6	289.8	289.4	289.3

The group optimum is `h = 1`. The deeper the population thinks, the **smaller** it is.

Individual fitness. Invasion trial, 50/50 of `h = 3` and `h = 9`, no mutation. The frequency of `h = 9` rises monotonically across three seeds, to 0.91 / 0.87 / 0.85 over 4,000 ticks. The deep planner **displaces** the shallow one.

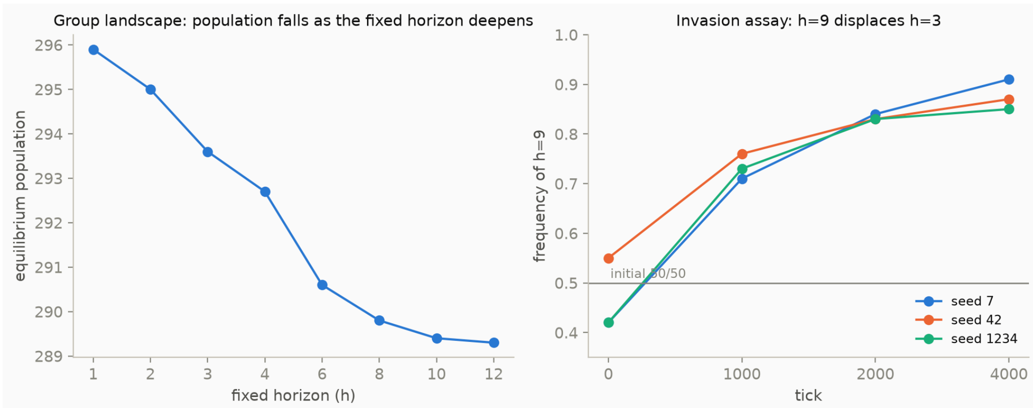

Figure 1. The two measurements side by side: group population falls as the fixed horizon deepens (left), while the deep planner `h=9` displaces the shallow one `h=3` in a direct invasion trial (right). The mismatch between the two **is** the finding.

The mismatch between the two measures **is** the finding, and it has the classic shape: thinking deep is individually advantageous and collectively bad. This is also why zeroing out competition makes evolved horizons **shallower** — competition does not restrain thinking, it **feeds** it. The gain from looking deep lies in **winning the neighbor's contested cell**, not in growing the pie.

4. The tournament, and an ESS that wasn't (the second layer)

The 3×9 pair is suggestive, not an evolutionarily stable strategy (ESS; Maynard Smith & Price, 1973; Maynard Smith, 1982). We ran the full tournament: **66 pairs × 8 seeds × 6,000 ticks**, 50/50 population, **only the horizon varying** (discount pinned at 0.80, otherwise the §1 compensation contaminates the contrast).

Dominance is **transitive and perfect**: each horizon beats exactly one more duel than the previous one; `h = 12` beats all 11, `h = 1` loses all 11. No cycle, no hidden mixed strategy. And the margin **saturates**: from total annihilation (1.000 against `h = 1`) to near coin-flip (0.52–0.56 at the top) — but it never turns non-positive, and so **the ESS is the ceiling** `h = 12`, censored by the world's maximum. A race with no endogenous brake.

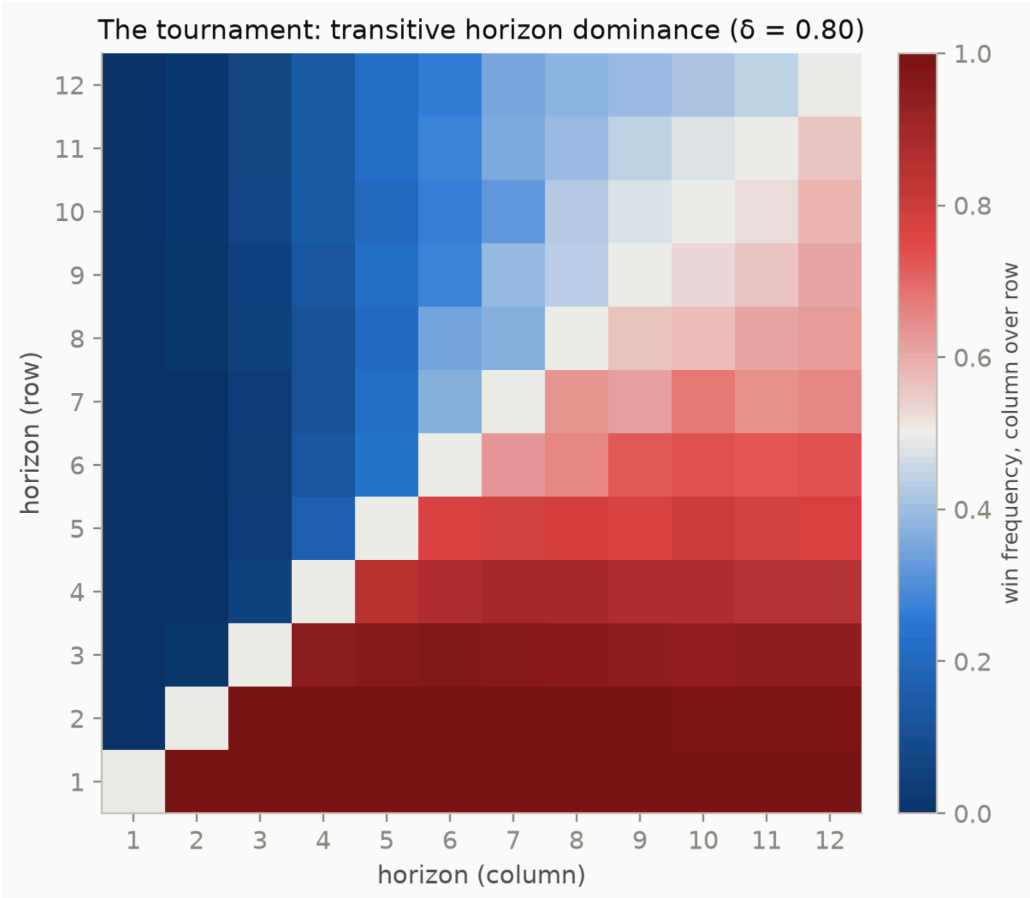

Figure 2. The full matrix of the 66 pairs. Dark blue in the bottom-left corner, dark red in the top-right: transitive and perfect dominance — every larger horizon beats every smaller one — with the margin saturating (whitening) near the ceiling `h = 12`.

We read this saturation as the **ceiling of effective depth** `min(h, 1/(1−δ))`: with δ = 0.80, `1/(1−0.80) = 5`, and the curve's knee fell at `h ≈ 4–6`. The **fixation → polymorphism** transition (exclusion below, coexistence above) fell at `h ≈ 4` — "exactly at the discount ceiling," we wrote.

This was wrong, and the error was a one-point error. Effective depth tied the whole argument together and had been measured at a single δ; at δ = 0.80, `1/(1−δ) = 5` — and `h ≈ 5` is also the **midpoint** of the 1..12 range. From one point, no one can distinguish "the knee is the discount ceiling" from "the knee just happened to land in the middle of the ruler."

We swept δ ∈ {0.30, 0.50, 0.80, 0.90, 0.95} (ceiling from 1.4 to 20), 15 pairs × 8 seeds, with the same tournament binary — the δ = 0.80 slice reproduces the original tournament in **120/120 rows**, which anchors the two measurements together. Three results:

(a) The knee moves — until it stops moving. The last step with a significant advantage goes from `h ≈ 2` (ceiling 1.4) to `h ≈ 3` (ceiling 2), `h ≈ 7` (ceiling 5), and `h ≈ 9` (ceiling 10). Censoring shows up where it should. But at δ = 0.95 (ceiling 20) it **comes back** to `h ≈ 6`. `1/(1−δ)` is a **low-δ approximation**: it describes when extra depth is *invisible*, and cannot tell when it is *harmful*.

(b) The transition does not move. It sits at `h = 4` for δ = 0.50, 0.80, 0.90, **and** 0.95, while the ceiling runs from 2 to
20. We had conflated **two scales** that only δ = 0.80 happens to

align: the knee of the margin, governed by the bottom of the ladder (where discount has leverage), and the transition, governed by the top — where the *relative* increment in depth is large (1→2 doubles; 11→12 adds 9%) and discount has no leverage at all.

(c) ESS = ceiling was an artifact of 0.80. At δ = 0.95 the ladder **inverts**: the last three rungs score 0.344, 0.240, and 0.202 (`t` = −5.5, −13, −14). **The shallow one wins.** Long-range duels — an independent measurement with non-adjacent pairs — agree: `h = 9` beats `h = 12` (`t = −11`), `h = 6` beats `h = 12` (`t = −4.4`). **There is an interior singular strategy** (Geritz, Kisdi, Meszéna, & Metz, 1998). And the optimum **falls** as δ rises: ≥ 12 at 0.80; ~10 at 0.90; ~7–8 at 0.95.

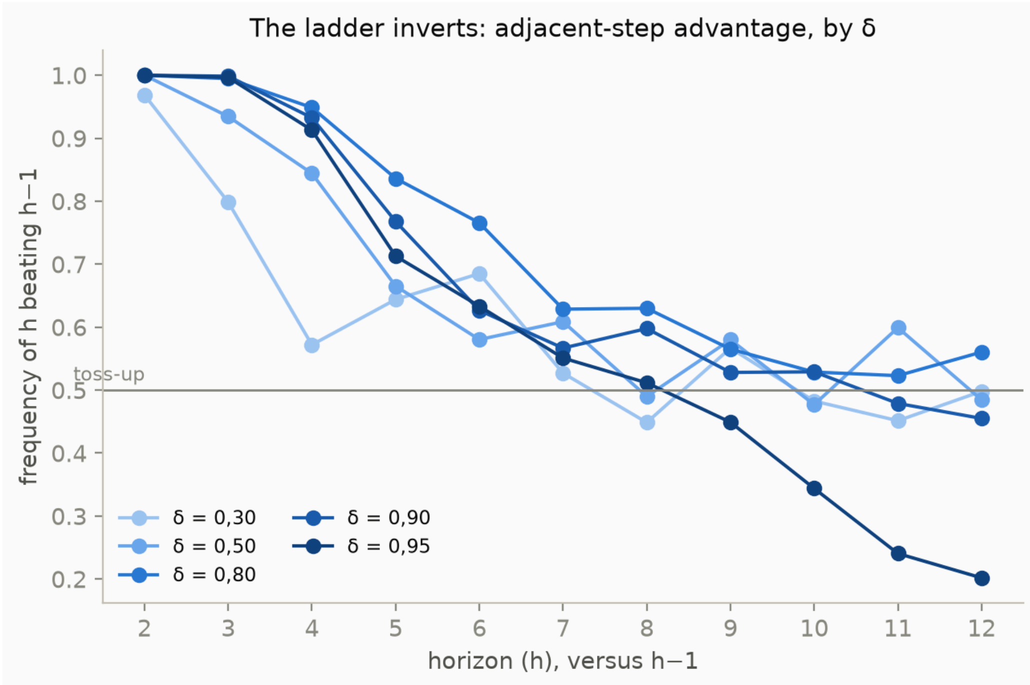

Figure 3. The advantage of each adjacent step (`h` versus `h−1`), one line per δ. At low δ (light blue) the line collapses early toward a toss-up; at δ = 0.95 (dark blue) it **crosses below 0.5** at the higher horizons — the extra step stops paying off, and the ladder inverts.

5. The brake is noise, not position (the third layer)

The inversion admitted two readings, and they make opposite predictions:

- **Noise** — the deep prediction is worse *against the world*. The deficit is **absolute** and shows up even with no rival of another type.
- **Stubbornness / positional** — the prediction is correct; the deep planner only loses because a shallow one arrives first at the food it planned for. Without a rival, the deficit **vanishes**.

The discriminant is a distinction this project already carries (*equilibrium population is a proxy for the group; for individual fitness, an invasion trial*): run **single-type** populations — every block with the same `h` and the same δ — and see whether the *duel's* inversion has a **solitary** counterpart.

It does. **Paired** difference by seed (the same seed across every `h`), 8 seeds:

δ	`pop(h=12) − pop(h=8)`	`standing_food(h=12) − standing_food(h=8)`
0.80	+2.04 ± 1.25 (`t = +1.6`)	+2.54 (`t = +2.1`)
0.90	−1.08 ± 1.25 (`t = −0.9`)	+13.55 (`t = +8.3`)
0.95	**−5.51 ± 1.27 (`t = −4.3`)**	**+31.45 (`t = +14.9`)**

At δ = 0.95, an entire world of `h = 12` blocks sustains **5.5 fewer blocks** than the same world with `h = 8` — and **there is no `h = 9` around to take its food.** The deficit is not positional: it is against the world. And the material trace is in the food **no one harvested.**

The dose-response is the signature. Standing food, paired against `h = 1`:

h	δ=0.80	δ=0.90	δ=0.95
2	**−2.3** `t−4.3`	**−2.5** `t−3.8`	**−2.2** `t−3.1`
6	+11.0 `t+7.9`	+22.9 `t+10.2`	+27.6 `t+13.7`
12	+20.3 `t+12.5`	+45.2 `t+14.4`	**+73.6** `t+17.3`

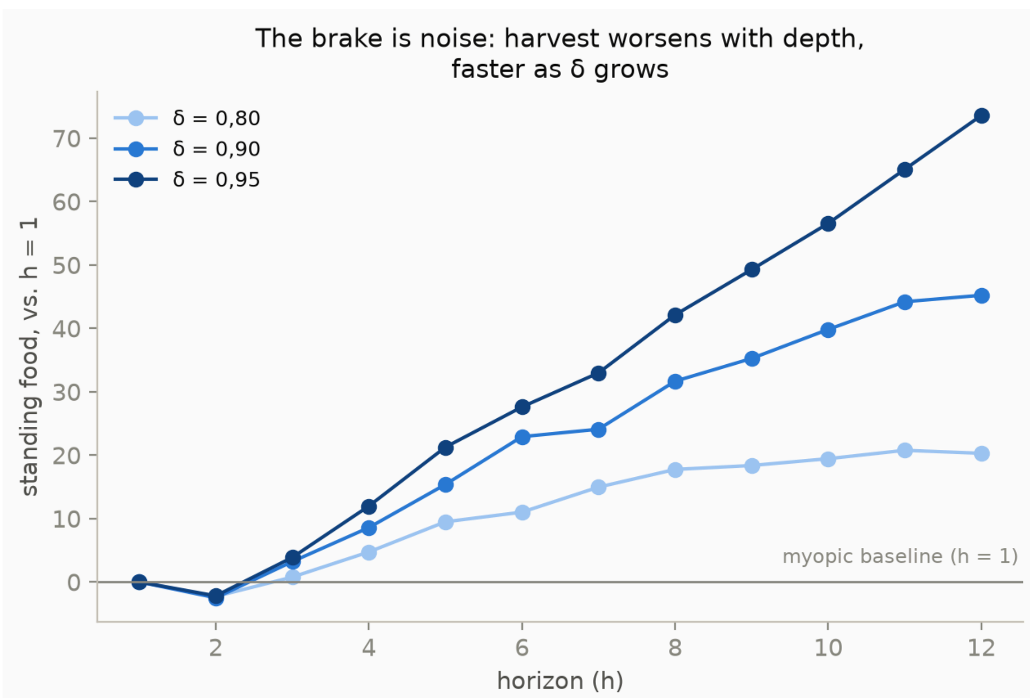

Figure 4. The full curve from `h = 1` to `12`, for the three δ. The harvesting peak (`h = 2`, the only dip below zero) is the same across all three; what changes is the slope after it — three times steeper at δ = 0.95 (dark blue) than at δ = 0.80 (light blue).

Three readings, and the third is the thesis:

The harvesting peak is `h = 2`. It is the only step that harvests better than the myopic one, at all three δ. One planning step pays off. **From `h = 3` onward, every additional step worsens the harvest**, monotonically, without exception.

The damage exists at every δ — including 0.80, where depth does not cost population (`t = +1.6`). In other words: in this world, **planning deep never harvested better.**

And discounting controls how much. The same `h = 12` leaves +20.3 / +45.2 / **+73.6** as δ goes to 0.80 / 0.90 / 0.95: the damage **more than triples**. The tail's error is the same across all three — same world, same depth. What changes is δ^k, **the weight the decision gives it**. At δ = 0.80, $0.8^{11} \approx 0.09$: the wrong tail enters discounted almost to nothing. At δ = 0.95, $0.95^{11} \approx 0.57$: it enters with more than half its weight, and costs 5.5 blocks.

> **Discounting is a regularizer.** It is not impatience: it is the block refusing to trust a prediction it cannot make.

This dissolves the counter-intuitive result of §4(c) without paradox: the more weight is put on the distant future, the less far one can look without being poisoned by one's own prediction. And it reinterprets the compensation of §1: the block that declares horizon 11 and discount 0.66 **is not gaming the metric** — it is protecting itself from its own hallucination. Lowering the discount is adaptive because it **regularizes**.

6. The synthesis: evolution lands past the productive optimum, and the surplus is position

Three independent measurements meet at one number:

quantity	value	source
harvesting optimum	`h ≈ 2`	single-type (§5)
group optimum (population)	`h = 1`	group landscape (§3)
evolved effective depth, no tax	**3.31 ± 0.24**	free evolution (§7)

Evolution lands **past** the productive optimum — by about a step and a half — and this surplus is **exactly the positional good**: depth that harvests nothing and exists only to win the neighbor's cell. The Red Queen, measured in steps.

And now the boundary. The horizon is a positional good **where discounting protects it** (δ ≲ 0.90): there, the extra depth is *harmless* — it does not cost population — and still **wins duels**. This is the definition of a positional good: it produces nothing, and one still needs it to avoid being displaced. Above that (δ = 0.95), the extra depth is *absolutely* bad, and there it is not an expensive good: it is a **mistake**, and selection corrects it. This correction is what produces the interior singular strategy of §4(c).

7. The tax, and why aligning the choice is not the same as restoring welfare

If the deep planner imposes on the commons a cost it does not pay, the classic remedy is to internalize the externality: `METABOLISM + c · depth`. A **Pigouvian tax on cognition** (Pigou, 1920).

We swept `c` across 7 values × 8 seeds × 30,000 ticks, with horizon and discount **free** (only the metabolic bill changes). The tax is charged on **effective depth**, not on declared horizon — otherwise the block would escape by lowering its discount, paying for steps that no longer carry weight. The choice of what to tax is not a detail: it is the difference between an instrument that bites and one that slips, and the evidence justifying it is the non-identifiability of §1 (`horizon_mean` has an SD up to **15×** that of effective depth; only when the tax is strong enough to pin everything at `h = 1` do the two readings converge).

`c`	effective depth	population
0	**3.31 ± 0.24**	284.4
0.04	1.82 ± 0.23	230.7
0.15	**1.04 ± 0.02**	186.9
0.30	1.01 ± 0.01	133.0

Evolved depth falls monotonically and **lands on the group optimum** (`h = 1`) at `c ≈ 0.15`. The predicted coincidence — a `c` at which individual and collective interest align — happened without tuning: it falls exactly where depth hits the `h = 1` that §3 had already measured, independently. The Pigouvian mechanism **works**, in the precise sense in which it was asked to.

And the price was not in the budget. Population falls monotonically with the tax: the `c` that aligns the choice has already cost **~35% of the population** (284 → 187). In a textbook Pigouvian tax the revenue is **redistributed**; here it is **burned** — extra metabolism that vanishes. So the tax does not correct the externality for free: it corrects **by digging an energy hole**. The positional externality §3 measures is small (~2% of population between `h = 1` and `h = 12`); the “cure” costs an order of magnitude more than the “disease.”

> **Aligning the choice is not the same as restoring welfare**, when the instrument of alignment is itself destructive. Whoever pays for coordination, pays.

And there is now an irony that §5 adds: **this world already had a brake**, and it is free. The exogenous, burned tax competes with an endogenous mechanism — the imprecision of the prediction itself — that produces an interior singular strategy without charging anyone anything.

> **[Addendum, 2026-07-26, note 21: the central conclusion of this paragraph was corrected — “aligning ≠ restoring welfare” was a property of burning the revenue, not of the Pigouvian tax itself.]** A per-capita dividend (the tax pool returned equally to every survivor, every tick) preserves the alignment almost untouched — evolved depth falls to the same place, within 0.16 of the burned version at every `c` (R1) — and **restores welfare beyond the baseline**: at the `c = 0.15` that aligns, the burned population is 66% of the no-tax baseline; the recycled one is **102%** (R2/R3). The pre-registered caveat — that redistributing to survivors near the reproduction ceiling would leave a residual cost — was **falsified in the favorable direction**: the residual is a surplus of ~2% at every `c ≥ 0.01`, of the same order as the positional externality §3 measures between `h = 1` and `h = 12` (a candidate reading, not experimentally isolated). The irony in this paragraph also changes shape: the two brakes — the endogenous one (§5, prediction imprecision) and the recycled exogenous one (note 21) — do not compete over who charges less; **both are already free**, through different mechanisms. `datasets/reciclagem.csv`, `papers/notes/21-o-imposto-que-recicla.md`.]**

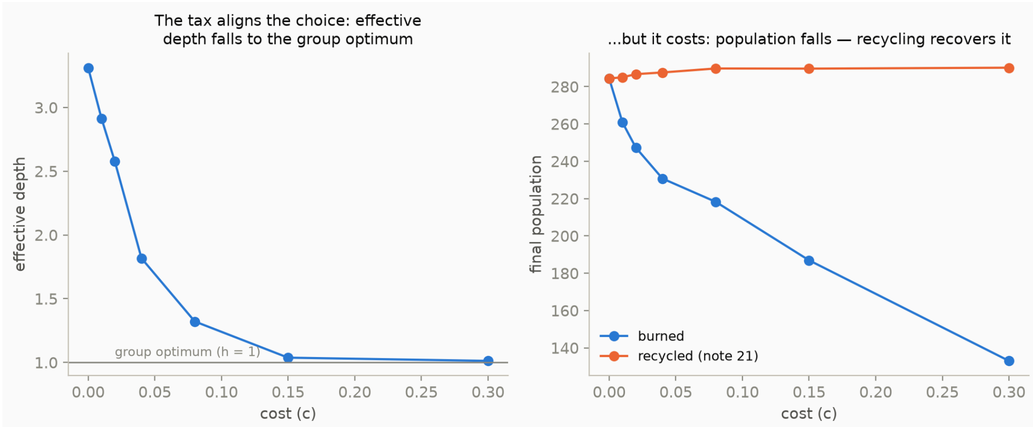

Figure 5. Left, effective depth falls monotonically and hits `h = 1` near `c = 0.15`. Right, the price of that correction: burned population (blue) falls throughout; recycled (orange, note 21) sits **above** the baseline instead of below it.

8. Threats to validity

- **The missing link.** We proved that the deep type harvests worse and that the damage scales with δ. We did **not** directly measure the k-step prediction error. §5 is inference to the best explanation, not a measurement of the link — closing it requires a probe comparing `predict_value` step by step against the realized outcome, and none existed. **[Addendum, 2026-07-20, note 18: the probe was built and the link closed — shorter than this paragraph feared. Only the plan's first step carries information (corr($\hat{g}_0$, r_0) = 0.60–0.73; corr ≤ 0.19 from k=1 onward — k* = 0 in every populated condition, 8/8 seeds), the premise of persistence dies at the very first replanning, and the tail's error *worsens* with δ (anticorrelation at δ=0.95): the inference of §5 becomes measurement, with a second channel — deep behavior fabricates part of its own noise. Remaining boundary: the probe measures the plan as *prediction*, not as *ranking* among candidate cells (`datasets/sonda-erro.csv`).] [Addendum, 2026-07-26, note 22: the remaining boundary was closed — the ordinal probe. The plan fails as a prediction (k* = 0, note 18) but ORDERS the up-to-9 candidate cells far better than chance: pairwise discordance between the predicted rank (`predict_value` alone) and the rank of what was actually extracted from each cell, by whoever, sits at 0.17–0.20 (chance = 0.5) in every populated condition, and the argmax hit rate is 3.8–4.5× chance — 8/8 seeds, 100.000% exact reconstruction of the real decision. Discordance barely worsens from `h = 4` to `h = 12` (|Δ| ≤ 0.015, against the collapse of `corr(g_k,r_k)` from note 18 already at**

`k = 1`): a candidate mechanism for the harvesting peak `h = 2` from note 17 — the RANK does not degrade the way the VALUE degrades. And the full decision (`u`, with the space term) tracks the realized outcome *better* when the block is hungry than when it is sated (Δ = 0.02–0.11 in favor, 8/8 seeds, `t` up to 29) — space recedes exactly when it should, by design, not by failure. Algebraic control: in the hermit (no rivals perceived), `space` is constant across candidates and `rank(u) ≡ rank(m)` by construction (an affine map, positive slope) — and the same comparison comes out inverted there (`t = −29`), as the algebra requires. `datasets/sonda-ordinal.csv`, `papers/notes/22-a-sonda-ordinal.md`.]

- **Standing food is a property of the world, not of the block.** More standing food is evidence of worse harvesting, but it is aggregate: it does not distinguish "every block harvests less" from "blocks cluster and leave regions untouched." Both are forms of worse harvesting; the second would be a spatial story, not a temporal-noise one.
- **8 seeds** in the tournaments and single-type runs; **3 seeds** in Phase 3 (§3). The effects that decide come out at `t` of 4 to 17; the nulls are *measured* nulls (`t = +1.6`), not "we cannot tell" — except at δ = 0.30, where the SD is 0.2–0.4 and the plateau is honestly "not distinguishable from 0.5 with 8 seeds."
- **Single-type ≠ hermit.** The blocks still see each other and contest cells; what does not exist is a block of a **different `h`**.
- **A 5-point discount grid.** The inversion lives between 0.90 and 0.95; exactly where the optimum leaves the ceiling, this grid cannot say. **[Addendum, note 19: the fine grid δ∈[0.88,0.96] answered this — `h*(δ)` falls 12→10→9→8→8→7 while `1/(1−δ)` rises 8.3→25 (opposite directions), and the ceiling is abandoned between δ=0.88 and 0.90 (`datasets/grade-fina.csv`).]**
- **50/50, not rare-invader.** With an interior singular strategy, the distinction started to matter for real — far more than it mattered when the ESS was the ceiling. **[Addendum, note 20: the rare-invader trial ran, and it renamed the finding — the interior point is NOT an ESS. Rare invaders from both sides grow (it is not uninvadable), but the rare `h*` invades both residents (it is convergence-stable): the textbook signature of an evolutionary branching point (Geritz et al., 1998). The "interior ESS" of §4/§6 above is, more precisely, the center of a protected polymorphism — which also gives a second, structural cause for the dispersed `horizon_mean` of note 15. Wherever the text says "interior ESS," read "convergence-stable, invasible interior singular strategy" (`datasets/invasor-raro.csv`). Direct test still missing: does free**

evolution produce a bimodal distribution of horizons?] [Addendum, 2026-07-26, note 23: the direct test ran, with a two-layer answer. In the SAME pinned background as this note (only the horizon free to mutate over 1..12), the branching point turns into REAL polymorphism — 8/8 seeds bimodal (mass ≥15% on both sides of `h*`) from δ=0.90 onward, full saturation (δ=0.80 gave 4/8, more ambiguous than expected from the paired dominance of notes 14/16). But in the REAL evolutionary run (urgency/space_weight/strategy also free), NO seed is bimodal — every run converges to a single peak, and it is the peak that disperses across seeds (mode 3 to 9, `sd=2.12`). The co-evolution of the other three traits does not erase the structural point — it erases *visible* bimodality, replacing it with path-dependence: each realization commits, early, to a different point of the same possibility space, never splitting the bet within a single run. It is the mechanism, now seen acting, behind the dispersed `horizon_mean` note 15 measured as a symptom. `datasets/bimodalidade.csv`, `papers/notes/23-o-teste-de-bimodalidade.md`.] [Addendum, 2026-07-26, note 26: the "δ=0.80 gave 4/8, more ambiguous" above got resolved. A classifier that DISCOVERS the valley on its own (comparing each bin against the running maximum on each side, without being told `h*`) re-analyzes the same histograms and votes 0/8 at δ=0.80 — what the fixed partition read as "mass on both sides" is a tail skewed toward the ceiling `h=12`, not two peaks separated by a valley. It reads in favor of mutation load, not real branching, at δ=0.80 — and confirms δ≥0.90 by an overwhelming majority (6-7/8), including discovering the valley exactly at the theoretical `h*` (δ=0.95, seed 1) without being told. `papers/notes/26-duas-linhagens-sem-h-conhecido.md`.] [Addendum, 2026-07-26, note 28: the "mutation load" verdict above was ISOLATED, not just voted on — and the answer is MIXED, not the single verdict note 26's addendum suggested. Turning off horizon mutation (only the initial 1..12 seeding supplies variation), 6/8 seeds collapse to near-monomorphism at the ceiling (low-side mass falls to 0.1–3.3%) — mutation load confirmed in those. But 2/8 do the opposite of what was predicted: with no new mutants at all, the low-side mass grows (0.39→0.58; 0.27→0.51) and the valley gets cleaner, not shallower (one seed reaches zero mass in the valley, across 20 samples over 5,000 ticks) — the signature of genuine polymorphism in a fraction of realizations, not mutational noise. δ=0.80 has no single verdict: it is a historically loaded boundary — which basin (pure ceiling or coexistence) a population visits

depends on its initial trajectory, not on δ alone. `datasets/mutacao-off.csv`, `papers/notes/28-mutacao-ou-ramificacao-em-delta-080.md`.]

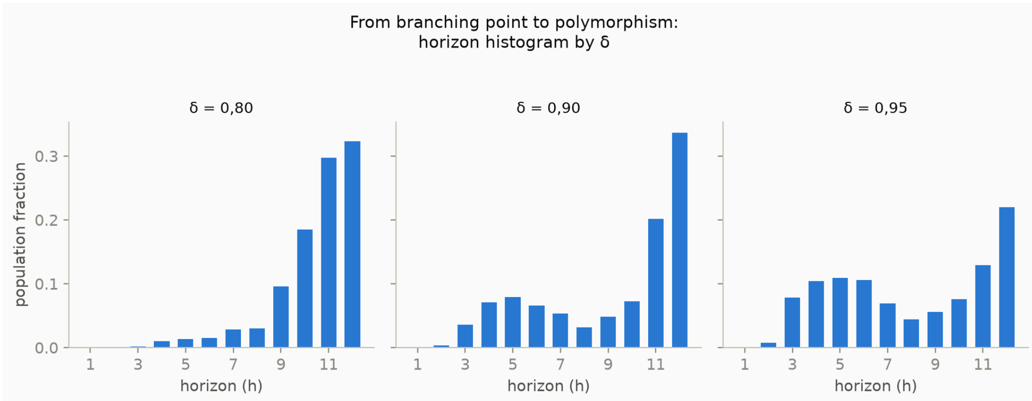

Figure 6. The full histogram behind the notes 26/28 verdict. At δ = 0.80 the mass is a tail skewed toward the ceiling — no valley. At δ = 0.90 and 0.95 both sides of `h*` gain mass of their own: the branching point turned into visible polymorphism.

- **`HORIZONTE_MAX = 12` censors** the top at low and medium δ.
- **One world.** "Depth produces nothing" is a property of *this* world — procedural food patches, a 3×3 neighborhood, one step per tick. A world with a more predictable reward structure should move the harvesting optimum upward, and that is an experiment, not an opinion.

9. What this means, if it means anything

The narrative "evolution learns to plan farther ahead" is, here, three things at once. It is **ill-formed** (the horizon is not identifiable without the discount). It is **true in a narrow regime** (one planning step pays off; the second already does not). And, where it is true, **it is not good news**: the depth evolution adds past the second step does not grow the pie — it transfers. Every block has to think deeper just to stay in place, and it comes at a cost for everyone.

What the 56 KB world adds to the Red Queen commonplace is the **brake**. It did not come from a tax, nor from a metabolic cost, nor from a design ratchet. It came from the product **rotting with distance**: the deep plan is a prediction the world does not honor, and discounting is the only organ that decides how much of that lie enters the decision. An arms race for a good that does not exist is self-limiting — not out of virtue, but because at some point the weapon starts firing backward.

If this holds for the 56 KB creature, the question is worth asking for the others.

Appendix — reproduction

There is **one** canonical `main.c`; every variant is a patch on a temporary copy, and every script accepts an alternate `main.c` (`git show <commit>:main.c`). Every claim in this paper has a note, a script, and a dataset:

section	note	script	dataset
§1, §3 (positional good, compensation)	Phase 3 of `ROADMAP.md`	—	—
§4 (tournament, ESS at δ=0.80)	14	`14-torneio.sh`	`torneio.csv`
§4 (δ sweep, interior ESS)	16	`16-desconto.sh`	`desconto.csv`
§5, §6 (noise vs. stubbornness; harvesting)	17	`17-tipo-unico.sh`	`tipo-unico.csv`
§7 (Pigouvian tax)	15	`15-imposto.sh`	`imposto.csv`
§7 (addendum — per-capita dividend)	21	`21-reciclagem.sh`	`reciclagem.csv`
§8 (addendum — the ordinal probe)	22	`22-sonda-ordinal.sh`	`sonda-ordinal.csv`
§8 (addendum — the bimodality test)	23	`23-bimodalidade.sh`	`bimodalidade.csv`

The notes (in `papers/notes/`, Portuguese, dated) record **dead hypotheses** — three in Phase 3, and the entire mechanism of §4, which note 16 overturned after note 14 had published it. In a project whose thesis is about what a measure carries, a dead hypothesis is data. **The order in which the errors were made and corrected is on record in the git history, and each note's pre-registration was committed before its run.**

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